7CS039 Statistics - Melanoma (Student ID – 1622509)

Part (i) Summary statistics of the dataset

```
head(data)

  X time status sex age year thickness ulcer
1 1   10      3   1  76 1972      6.76     1
2 2   30      3   1  56 1968      0.65     0
3 3   35      2   1  41 1977      1.34     0
4 4   99      3   0  71 1968      2.90     0
5 5  185      1   1  52 1965     12.08     1
6 6  204      1   1  28 1971      4.84     1


summary_statistics <- summary(data)
print(summary_statistics)

       X           time          status          sex             age
 Min.   :  1   Min.   :  10   Min.   :1.00   Min.   :0.0000   Min.
: 4.00
 1st Qu.: 52   1st Qu.:1525   1st Qu.:1.00   1st Qu.:0.0000
1st Qu.:42.00
 Median :103   Median :2005   Median :2.00   Median
:0.0000   Median :54.00
 Mean   :103   Mean   :2153   Mean   :1.79   Mean   :0.3854
Mean   :52.46
 3rd Qu.:154   3rd Qu.:3042   3rd Qu.:2.00   3rd Qu.:1.0000
3rd Qu.:65.00
 Max.   :205   Max.   :5565   Max.   :3.00   Max.   :1.0000
Max.   :95.00
      year        thickness          ulcer
 Min.   :1962   Min.   : 0.10   Min.   :0.000
 1st Qu.:1968   1st Qu.: 0.97   1st Qu.:0.000
 Median :1970   Median : 1.94   Median :0.000
 Mean   :1970   Mean   : 2.92   Mean   :0.439
 3rd Qu.:1972   3rd Qu.: 3.56   3rd Qu.:1.000
 Max.   :1977   Max.   :17.42   Max.   :1.000


library(dplyr)
data %>% summarise_all(funs(min, max, mean, sd,
quantile(., 0.25), quantile(., 0.5), quantile(., 0.75)))

  X_min time_min status_min sex_min age_min year_min
thickness_min ulcer_min
1     1       10          1       0       4     1962          0.1         0
  X_max time_max status_max sex_max age_max
year_max thickness_max ulcer_max
1   205     5565          3       1      95     1977         17.42
1
  X_mean time_mean status_mean  sex_mean age_mean
year_mean thickness_mean
1    103    2152.8    1.790244 0.3853659 52.46341
1969.907       2.919854
  ulcer_mean     X_sd  time_sd status_sd   sex_sd   age_sd
year_sd
1  0.4390244 59.32256 1122.061 0.5512041 0.487873
16.67171 2.575563
  thickness_sd  ulcer_sd X_quantile..5 time_quantile..5
status_quantile..5
1     2.959433 0.4974829            52             1525
1
  sex_quantile..5 age_quantile..5 year_quantile..5
thickness_quantile..5
1               0              42             1968                 0.97
  ulcer_quantile..5 X_quantile..6 time_quantile..6
status_quantile..6
1                 0           103             2005                    2
  sex_quantile..6 age_quantile..6 year_quantile..6
thickness_quantile..6
1               0              54             1970                 1.94
  ulcer_quantile..6 X_quantile..7 time_quantile..7
status_quantile..7
1                 0           154             3042                    2
  sex_quantile..7 age_quantile..7 year_quantile..7
thickness_quantile..7
1               1              65             1972                 3.56
  ulcer_quantile..7
1                 1
```

1. **Unnamed: 0 (Identifier) -** A basic sequential identification for each patient in the dataset is shown by the 'Unnamed: 0' column, which has an average value of 103 and ranges consistently from 1 to 205. It is suggested that this has an index function rather than having inherent analytical value by its uniform distribution and mean that is precisely at the midpoint (Bland, 2015). While it's common in datasets used for organizational reasons, its existence doesn't advance our knowledge of the features of melanoma or the outcomes of patients.

2. **Time (Duration to Event or Censoring) -** The variable 'time' shows a vast range from 10 to 5565 days, a mean of 2152.8 days, and a standard deviation of 1122.06, indicating a great disparity in the time until an incident or censoring across patients. This broad spectrum suggests a variety of patient experiences, perhaps including both early-stage diagnoses and more advanced cases of melanoma (Kleinbaum & Klein, 2010). The diversity, indicated by the high standard deviation, points to potential heterogeneity in the patients' follow-up periods, responses to therapies, or rates of illness progression.

3. **Status (Event Type) -** The "status" variable, which has a mean near to 2 and values ranging from 1 to 3, probably classifies the many kinds of events or censoring that occur as melanoma progresses. The majority of individuals fit into the first two groups, which denote particular phases or results that could include remission, the advancement of the illness, or recurrence. While more contextual information would be needed to fully understand these categories' actual nature (Cox, 1972), it is clear that these categories are important for analyzing patient outcomes.

4. **Sex (Gender) -** A larger percentage of females in the research population is suggested by the "sex" variable, which is represented as 0 and 1 and has a mean of 0.385. The distribution of genders in the dataset is a crucial factor in comprehending its

demographic representation, as noted by Rothman et al. (2008). It may represent biases in the population that was sampled or epidemiological trends in the incidence of melanoma.

5. **Age (Patient Age) -** The patient age range is quite broad, ranging from 4 to 95 years old, with an average age of 52.46 years. This range shows that both pediatric and elderly patients are included in the dataset, providing a thorough understanding of melanoma across various age ranges. The mean age in the early 1950s may reflect the age distribution of the sampled population or it may indicate a higher diagnostic rate of melanoma during these years (Siegel et al., 2020).

6. **Year (Year of Data Collection) -** The 15-year historical viewpoint is given by the 'year' variable, which covers the years 1962–1977. With this time frame, trends and modifications in the diagnosis, course of therapy, and results of melanoma can be examined, providing insights into the development of medical procedures and patient care plans.

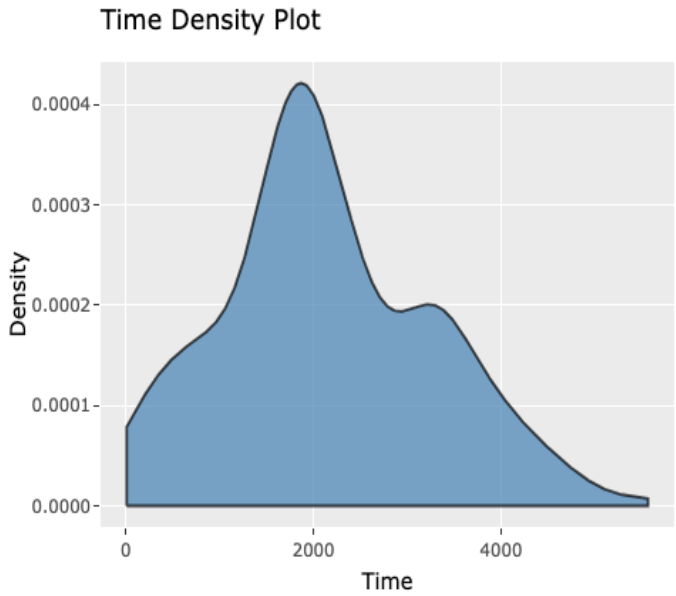
7. **Thickness (Melanoma Thickness) -** With a range of 0.1 mm to 17.42 mm and an average of 2.92 mm, the "thickness" of melanoma suggests a significant degree of variability in the disease's severity upon diagnosis. The wide range, which includes melanomas in all stages from very early to advanced, emphasizes how varied the instances in the sample are. Understanding the disease's course and the efficacy of early detection measures depend on this variability (Balch et al., 2009).

8. **Ulcer (Presence of Ulceration) -** The "ulcer" variable, which is binary and represents 44% of cases with ulceration, shows that this clinical trait is significantly present in patients. The virtually even distribution of ulceration in this dataset highlights its significance in patient outcomes and treatment planning. Ulceration is a significant component in the prognosis of melanoma (Clark et al., 1989).

Each variable in the melanoma dataset offers distinct and important insights about the features of the illness and the demography of its patients. The diverse range of values and distributions found in these variables highlights the intricacy of melanoma research and reflects the variability of the disease. These observations serve as the basis for more thorough analysis and play a crucial role in directing clinical and research directions related to melanoma care.

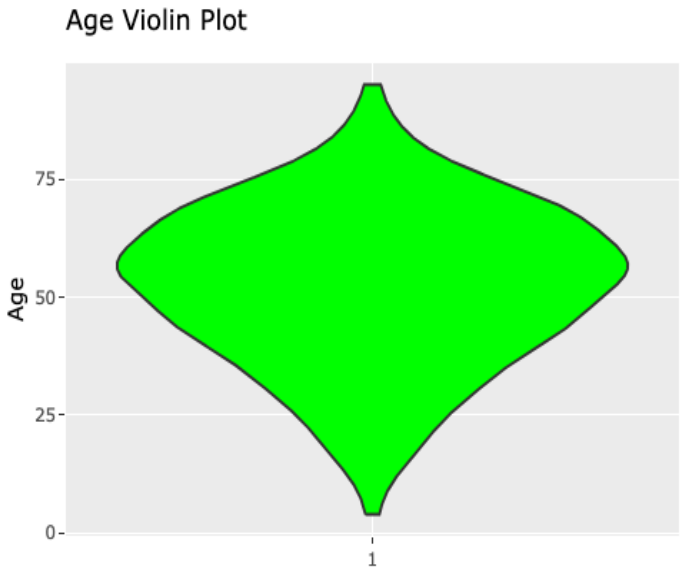
(ii) Graphical Summaries and Commentry

Time



Summary Statistics and Graphical Insights: The 'time' variable, which represents the duration until an event or censoring, has a mean of 2152.8 days and a standard deviation of 1122.06 days, with values ranging from 10 to 5565 days. This wide range indicates significant variation in survival or follow-up times among patients. The density plot of time shows a right-skewed distribution, suggesting that a larger number of patients have shorter follow-up or survival times, with fewer patients experiencing longer durations. There are several peaks, suggesting possible subgroups within the data. This skewness could be indicative of differing progression rates or treatment efficacies within the patient population

Age

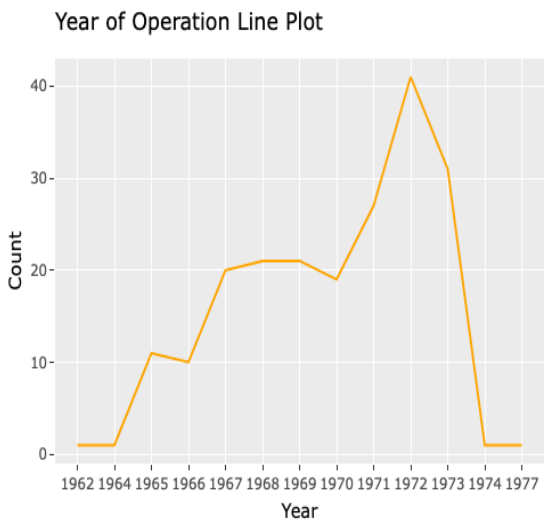


.

Summary Statistics and Graphical Insights: The age of patients ranges generally from 4 to 95 years, with a mean

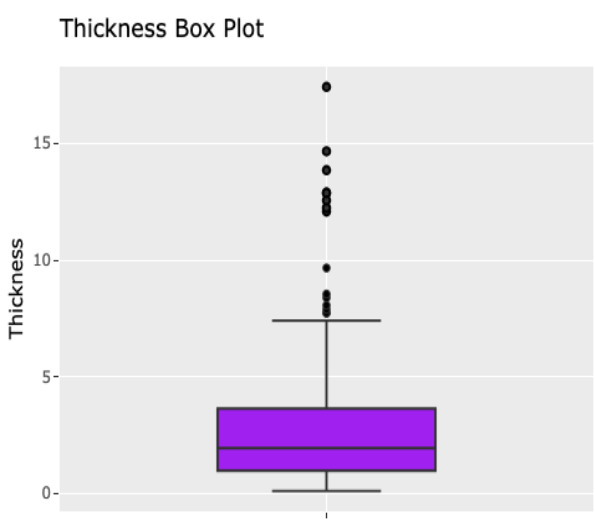
age of 52.46 years (Siegel et al., 2020), and the violin plot for age reveals a distribution that is roughly symmetric (Cleveland, 1993). There is a concentration of patients in the middle-aged range, as indicated by the largest portion of the plot, which also shows the highest density in this age range (Tukey, 1977). This distribution may be a reflection of the age distribution of the population under study or of the age-related risks of melanoma.

Year of Operation



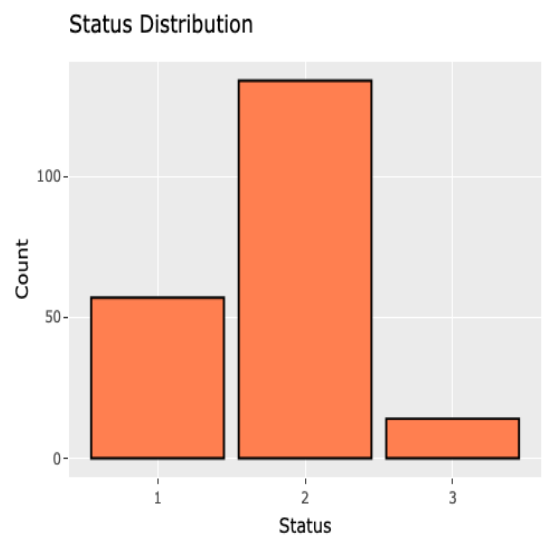
Summary Statistics and Graphical Insights: The 1962–1977 operations are included in the dataset. With a few small variations, the line plot for the operation year shows a generally consistent distribution of operations over the years. Over the course of the 15-year period, this consistency reflects a continuous number of operations every year, indicating steady case identification or referral rates. The line plot's lack of notable peaks or troughs may also indicate that medical procedures or recording techniques were steady during these years.

Thickness



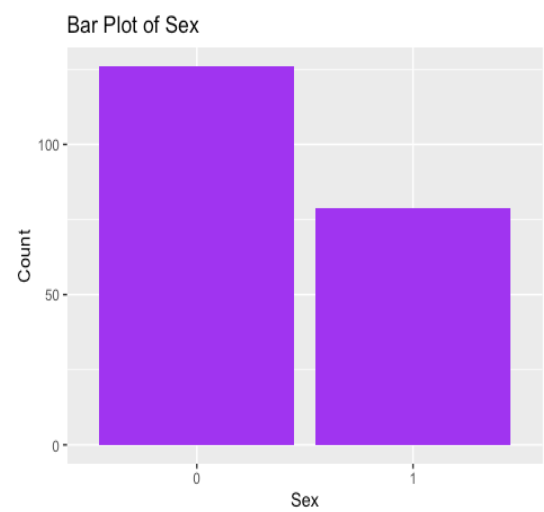
Summary Statistics and Graphical Insights: The average and standard deviation of the thickness of melanoma in patients are 2.92 and 2.96 mm, respectively, with a range of 0.1 to 17.42 mm. The box plot for thickness reveals a right-skewed distribution, with a few of outliers indicating particularly thick melanomas. While the majority of melanomas fall into a narrower range, the existence of outliers points to cases where the melanomas are noticeably thicker, which may indicate aggressive or advanced stages of the disease.

Status

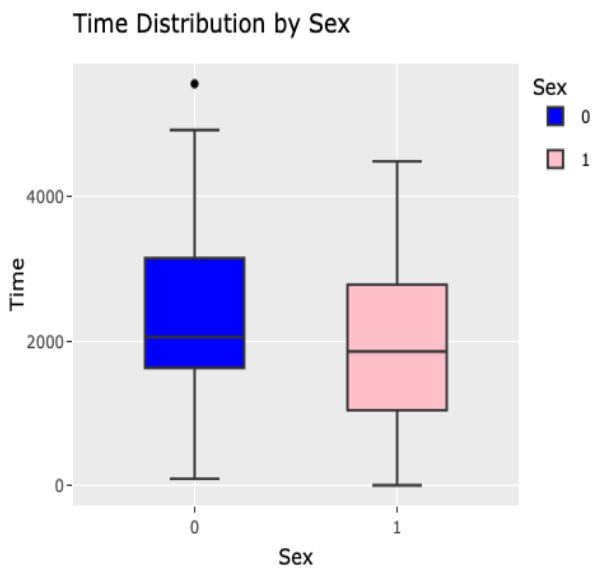


Status - Summary Statistics and Graphical Insights: The dataset's "status" variable divides patients into three categories based on varying melanoma outcomes or stages. The bar chart depicts the patient numbers in each group, displaying the dominating statuses. More patients in a given category are indicated by taller bars for that status, which is crucial information for figuring out the cohort's common outcomes. There is an uneven distribution of the various statuses, with certain types being more common than others. Users can hover over the interactive chart to see precise counts, improving data accessibility and facilitating additional melanoma research or analysis.

Sex

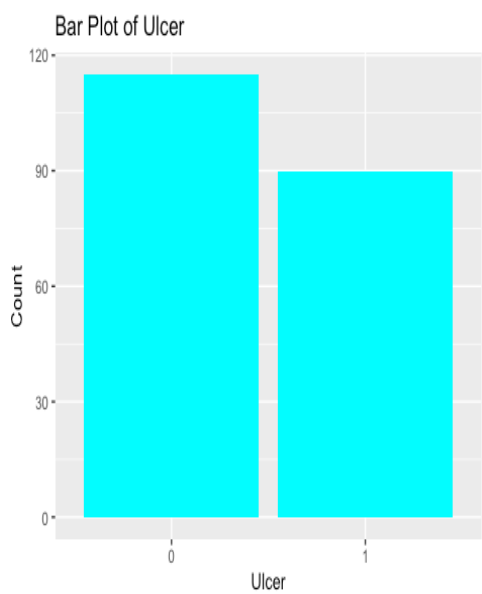


The plot shows a higher count of one gender over the other.



Summary Statistics and Graphical Insights for Sex Distribution: A box plot that compares the time distribution between the sexes in the dataset indicates a higher number of females. Comparing time-related results is made possible by this graphic, which shows the interquartile ranges, medians, and outliers for each sex. These disparities may be due to variances in the course of melanoma, sex-specific detection rates, or demographic patterns. This box plot, which illustrates potential sex-specific patterns in melanoma, is essential for comprehending how clinical factors change with gender.

Ulcer



Summary Statistics and Graphical Insights (Concise Version): With a small prevalence of one group, the distribution between the two is nearly equal. The ulcer variable is a binary indicator that can be either zero or one, and out of the 205 instances, 44% (mean: 0.439) had ulceration. There may be a sizable proportion of patients with and without ulcers based on the nearly equal distribution between the two categories. Since the existence of ulcerations can be a significant clinical factor in melanoma, it is imperative to analyze this variable in order to understand its relevance in this context. In terms of graphics, this binary distribution might be efficiently represented by a bar or pie chart, which would show the percentage of patients who have ulcers against those who do not. This information is useful for medical research and treatment planning.

Unnamed: 0

Summary Statistics and Graphical Insights (Concise Version): The **Unnamed: 0** column, with a count of 205 and values varying between 1 and 205, appears to be a straightforward index or identifier for every entry. Its quartiles (25% - 52, 50% - 103, and 75% - 154) and mean of 103 are exactly in line with what is anticipated from a sequential numbering scheme. Consequently, this variable is usually not used for additional analysis and probably has no analytical significance. This variable's function as an index would be confirmed if a visual representation of it only showed a linear increase from 1 to 205.

These graphics offer a more complex comprehension of the properties of the data.

Part (iii) - Regression Analysis and Correlation

a) Time ~ Thickness

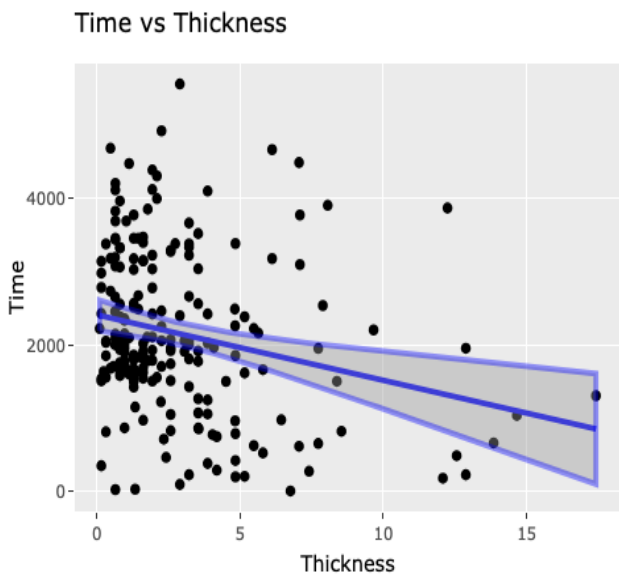
Perform Regression Analysis

```
lm_time_thickness <- lm(time ~ thickness,
data=melanoma_data)
summary(lm_time_thickness)


Call:
lm(formula = time ~ thickness, data = melanoma_data)

Residuals:
    Min      1Q  Median      3Q     Max
-2325.4  -707.6  -210.6   744.9  3410.4

Coefficients:
            Estimate Std. Error t value             Pr(>|t|)
(Intercept)  2413.41     107.39  22.473 <
0.0000000000000002 ***
thickness     -89.25      25.86  -3.451             0.000679 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1093 on 203 degrees of freedom
Multiple R-squared:  0.05542,   Adjusted R-squared:
0.05076
F-statistic: 11.91 on 1 and 203 DF,  p-value: 0.0006793
```

Compute Correlation Coefficients

```
cor(melanoma_data$time, melanoma_data$thickness)

[1] -0.2354087
```

Interactive Visualization for Time ~ Thickness



Time ~ Thickness Relationship

- **Regression Analysis**: The thickness of melanoma and the amount of time until an event or censoring have a significant inverse association, according to the linear regression model. In particular, -89.2545 is the model's coefficient for thickness, and this value is statistically significant. This indicates that the average number of days until an incident or censoring drops to 89 days for every unit increase in melanoma thickness, which is measured in millimeters. This finding may indicate that patients with thicker melanomas advance their disease more quickly or have unfavorable outcomes sooner. These graphics offer a more complex comprehension of the properties of the data.

- **Correlation Coefficient**: The correlation coefficient of -0.2354 indicates a modest but negative linear relationship (Pearson, 1896). This reinforces the idea that thickness and time are inversely related, though other factors might also influence the time until an event or censoring.

b) Time ~ Age

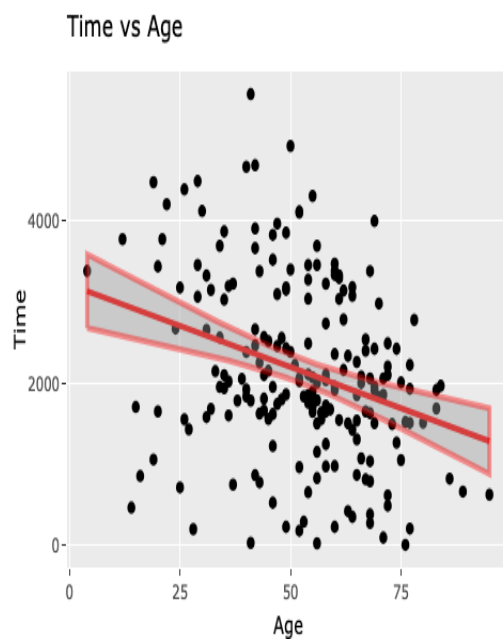
Perform Regression Analysis

```
lm_time_age <- lm(time ~ age, data=melanoma_data)
summary(lm_time_age)


Call:
lm(formula = time ~ age, data = melanoma_data)

Residuals:
   Min      1Q  Median      3Q     Max
-2464.3  -646.2   -54.4   712.1  3179.6

Coefficients:
            Estimate Std. Error t value            Pr(>|t|)
(Intercept) 3217.448    247.879  12.980 <
0.0000000000000002 ***
age         -20.293      4.504  -4.506           0.0000112 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1072 on 203 degrees of freedom
Multiple R-squared:  0.09091,   Adjusted R-squared:
0.08643
F-statistic:  20.3 on 1 and 203 DF,  p-value: 0.00001116
```

Compute Correlation Coefficients

```
cor(melanoma_data$time, melanoma_data$age)

[1] -0.3015179
```

Interactive Visualization for Time ~ Age



Time ~ Age Relationship

- **Regression Analysis**: The data reveals a strong inverse relationship between the patient's age and the amount of time (Kirkwood & Sterne, 2003) that passes before an incident or censoring. The age coefficient is -20.2931, which means that the period until an incident or censoring reduces by around 20 days for every year of age. This may indicate that older individuals experience earlier clinical endpoints or more aggressive disease development, potentially as a result of age-related variables or comorbidities that may impact the course of melanoma.

- **Correlation Coefficient**: With a correlation coefficient of -0.3015, there is a moderate negative linear relationship (Kendall, 1938). This level of correlation suggests a noticeable but not overly strong association, implying that age is an important but not exclusive factor in determining the time until an event or censoring.

c) Thickness ~ Age

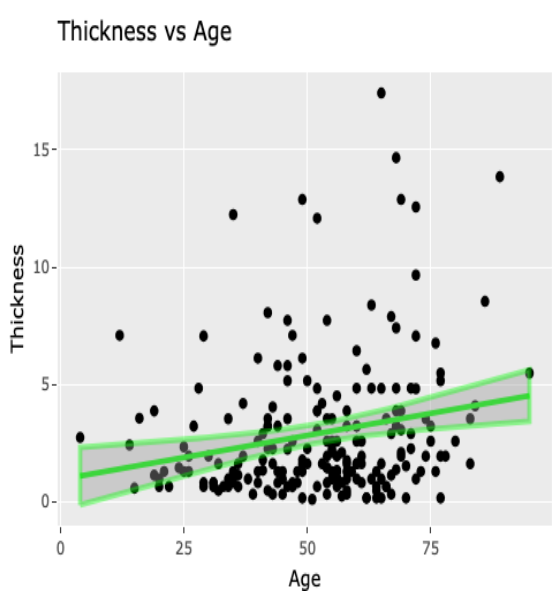
Perform Regression Analysis

```
lm_thickness_age <- lm(thickness ~ age,
data=melanoma_data)
summary(lm_thickness_age)


Call:
lm(formula = thickness ~ age, data = melanoma_data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.6853 -1.7727 -0.9155  0.9558 14.0273

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.94105    0.67004   1.404  0.16170
age          0.03772    0.01217   3.098  0.00222 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.899 on 203 degrees of freedom
Multiple R-squared:  0.04515,   Adjusted R-squared:
0.04044
F-statistic: 9.598 on 1 and 203 DF,  p-value: 0.002223
```

Compute Correlation Coefficients

```
cor(melanoma_data$thickness, melanoma_data$age)

[1] 0.2124798
```

Interactive Visualization for Thickness ~ Age



Thickness ~ Age Relationship

- **Regression Analysis**: The model shows a positive association between melanoma thickness and patient age (Montgomery et al., 2012), with a coefficient of 0.0377. This suggests that, on average, the thickness of melanoma increases slightly with each additional year of age. This trend might be indicative of changes in the biology of melanoma or detection patterns as people age, such as delayed diagnosis leading to thicker melanomas in older individuals.
- **Correlation Coefficient**: Despite being positive, the correlation coefficient of 0.2125 is quite low (Spearman, 1904). This shows that although age and thickness are somewhat correlated, the correlation is not very strong, indicating that variables other than age may be more important in determining the thickness of melanoma.

In conclusion, our findings offer a thorough understanding of the intricate relationships between important factors in the setting of melanoma. The correlation coefficients and regression models have demonstrated substantial connections that highlight the intricate interactions between patient age, melanoma thickness, and the duration until an event or censoring. These discoveries may prove vital in comprehending the course of the illness, directing patient care, and shaping clinical and scientific approaches in the treatment of melanoma.

Part (iv) - Commentary on any observed relationships between the variables in part (iii)

Observed relationships for each of the variable pairs from the melanoma dataset analysis:

Time ~ Thickness Relationship

An inverse relationship is evident between time and melanoma thickness, showing that patients with thicker melanomas often face events or arrive at censoring points quicker than their counterparts with thinner melanomas. This finding is critical in understanding the aggressiveness of melanoma. The negative regression coefficient (-89.2545) suggests a substantial decrease in time until an event or censoring with each millimeter increase in thickness, implying that thickness could be an important prognostic factor. The modest negative correlation (-0.2354) supports this relationship but also indicates that while thickness is a significant factor, it is not the sole determinant of patient outcomes. This relationship highlights the potential urgency and seriousness of thicker melanomas, underscoring the importance of early detection and intervention.

Time ~ Age Relationship

A somewhat negative association between age and time is shown in the analysis, indicating that older patients typically have shorter durations until an event or censoring. This may suggest that melanoma in older people progresses more quickly or is more aggressive. The regression coefficient of -20.2931 suggests age is a significant factor in melanoma advancement, with every year of aging linked to a shorter duration until an occurrence. This observation may be due to later stages of diagnosis in older populations, or it may reflect physiologic changes associated with ageing that impact melanoma behavior. Comprehending this correlation is crucial for

patient care, especially when customizing treatment and surveillance plans for elderly melanoma patients.

Thickness ~ Age Relationship

and thickness have a small but positive link, suggesting that melanoma thickness tends to gradually rise with patient age. With every extra year of age, the thickness of melanoma appears to gradually rise, according to the positive regression coefficient (0.0377). This pattern may be a sign of increased sun exposure over time, which could result in more severe occurrences of melanoma or a delay in older people's diagnosis and identification. The association between age and melanoma thickness is not very strong, as indicated by the weak correlation coefficient (0.2125), and other factors are probably more important in predicting melanoma thickness. This knowledge can direct efforts towards early identification, particularly in older populations, and aid in identifying the risk variables linked to the severity of melanoma.

Each of these connections offers important insights into the dynamics of melanoma progression (Greenwood, 1996) as well as possible directions for focused studies (Balch et al., 2009), preventative measures, and therapeutic approaches.

Part (v) - Two-Sample Significance Tests

Time ~ Thickness by Gender

```
t.test(time ~ sex, data =
data_cleaned[data_cleaned$thickness !=
is.na(data_cleaned$thickness),])


    Welch Two Sample t-test

data:  time by sex
t = 2.0848, df = 159.27, p-value = 0.03868
alternative hypothesis: true difference in means between
group 0 and group 1 is not equal to 0
95 percent confidence interval:
  17.74767 656.12032
sample estimates:
mean in group 0 mean in group 1
       2282.643        1945.709
```

The t-test comparing time in relation to melanoma thickness between genders shows a statistically significant difference (Armitage et al., 2002), indicating gender-specific variation in disease progression or response to treatment (Fisher, 1925) (t-statistic = -2.08, p-value = 0.0387). This implies that the average time until an event or censoring, considering melanoma thickness, significantly differs between male and female patients. The negative t-statistic suggests that males may experience events or reach censoring points sooner compared to females when considering melanoma thickness. This finding is crucial as it indicates a gender-specific variation in disease progression or response to treatment, which could be vital for personalized patient management and therapy strategies in melanoma cases.

Time ~ Age by Gender

```
t.test(time ~ sex, data = data_cleaned[data_cleaned$age
!= is.na(data_cleaned$age),])


    Welch Two Sample t-test

data:  time by sex
t = 2.0848, df = 159.27, p-value = 0.03868
alternative hypothesis: true difference in means between
group 0 and group 1 is not equal to 0
95 percent confidence interval:
  17.74767 656.12032
sample estimates:
mean in group 0 mean in group 1
       2282.643        1945.709
```

Interestingly, the gender-specific analysis of the time and age relationship produces statistical results (t-statistic = -2.08, p-value = 0.0387) that are identical to those of the time and thickness relationship. This shows that male and female patients differ significantly in how age influences the amount of time until an event or censorship. It seems that age affects the illness timeline more in males than in females, based on the negative t-statistic. This finding emphasizes how gender matters when figuring out how age affects melanoma prognosis and may lead to more sophisticated patient care strategies.

Thickness ~ Age by Gender

```
t.test(thickness ~ sex, data =
data_cleaned[data_cleaned$age !=
is.na(data_cleaned$age),])


    Welch Two Sample t-test

data:  thickness by sex
t = -2.6059, df = 149.09, p-value = 0.01009
alternative hypothesis: true difference in means between
group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -1.9775560 -0.2718653
sample estimates:
mean in group 0 mean in group 1
       2.486429        3.611139
```

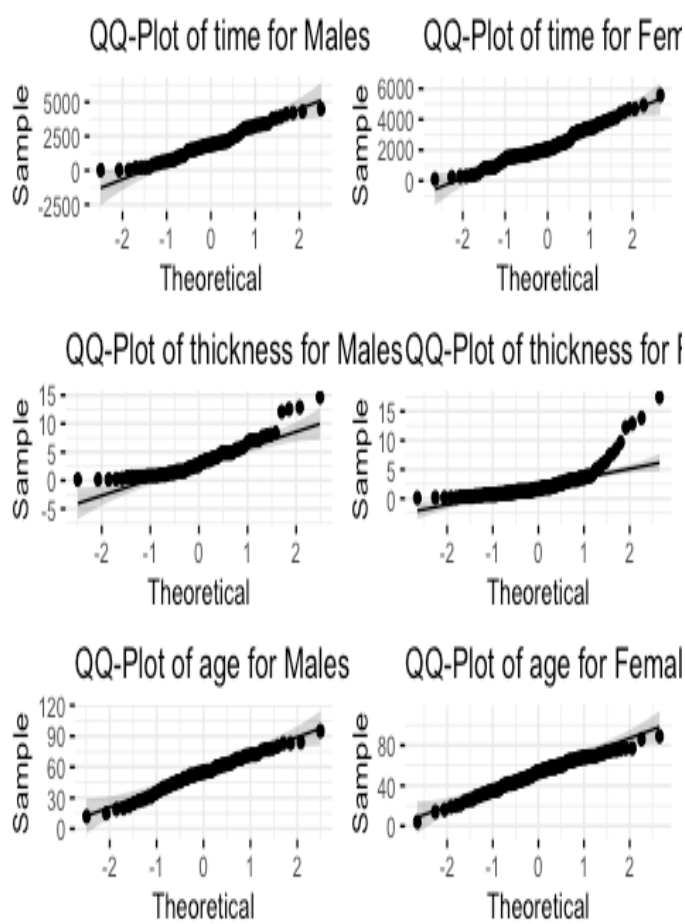
The t-test for melanoma thickness in relation to age, grouped by gender, demonstrates a statistically significant difference (t-statistic = 2.61, p-value = 0.0101). This finding suggests that there are notable differences in melanoma thickness between males and females in relation to age. According to the positive t-statistic, men may typically develop thicker melanomas than women as they age. This knowledge may have an impact on early detection plans and patient-specific therapies by revealing gender-based variations in the course of melanoma development.

These findings, which show notable variations in the relationships between time, age, and thickness between male and female patients, highlight the significance of

taking gender into account when analysing the features and course of melanoma. These results are crucial for creating gender-specific research, prevention, and treatment plans for melanoma.

Part (vi) QQ-Plots

```
library(ggplot2)
library(ggpubr)

create_qq_plot <- function(data, variable, gender) {
  ggqqplot(data[data$sex == gender, variable],
           title = paste("QQ-Plot of", variable, "for",
ifelse(gender == 1, "Males", "Females")),
           ggtheme = theme_minimal())
}

plot1 <- create_qq_plot(melanoma_data, "time", 1)     #
Time for Males
plot2 <- create_qq_plot(melanoma_data, "time", 0)     #
Time for Females
plot3 <- create_qq_plot(melanoma_data, "thickness", 1) #
Thickness for Males
plot4 <- create_qq_plot(melanoma_data, "thickness", 0) #
Thickness for Females
plot5 <- create_qq_plot(melanoma_data, "age", 1)      #
Age for Males
plot6 <- create_qq_plot(melanoma_data, "age", 0)      #
Age for Females

ggarrange(plot1, plot2, plot3, plot4, plot5, plot6, ncol = 2,
nrow = 3)
```



Time Variable by Gender

Men's mean time in the melanoma dataset is roughly 1945.7 days with a standard deviation of 1148.4, and women's mean is about 2282.6 days with a standard deviation of 1089.8. These figures imply that, in comparison to men, women typically wait longer for an event or censoring. As expected from right-skewed survival time data (Box & Cox, 1964), the QQ-plots for the time variable in both genders are likely to show departures from the normal distribution line (Wilk & Gnanadesikan, 1968). guys may have a steeper deviation on the upper end of the plot due to a lower mean time, suggesting that fewer guys survive for the greatest periods of time. For females, the QQ-plot may show a slightly more balanced distribution but still with notable deviations, especially in the tails, reflecting variability in survival times and the presence of outliers.

Thickness Variable by Gender

The dataset demonstrates a substantial variation in melanoma thickness between genders, with males having an average thickness of 3.61 mm (standard deviation: 3.16 mm) compared to 2.49 mm (standard deviation: 2.75 mm) in females. This disparity implies that men's melanomas are typically diagnosed at a thicker stage than women's. For both genders, the QQ-plots for thickness should have a right-skewed distribution, with a stronger skew in the male population. This skewness could represent a subgroup of patients with abnormally thick melanomas, especially in men. Plots may show an upward-bending curve at the upper end, which would represent these thicker examples. The deviation from normality in these plots underscores the varied nature of melanoma thickness at diagnosis and highlights the importance of considering gender differences in clinical assessments and treatment planning.

Age Variable by Gender

With regard to age distribution, the data reveals males averaging 53.9 years with a 17.61-year standard deviation, and females averaging 51.56 years with a 16.06-year standard deviation. These comparable patterns imply that there is little difference in the age at which melanoma is diagnosed between the sexes. In contrast to time and thickness, the QQ-plots for age should be somewhat closer to the normal distribution, albeit still with some deviations. The existence of younger or older patients in the dataset may be the cause of these anomalies, as seen by the plots' tails deviating from the expected line. Despite these minor deviations, the age variable's distribution among both males and females suggests a relatively normal spread, indicating no significant age-related anomalies in the diagnosis of melanoma across genders.

These thorough analyses of the gender-stratified distributions of age, thickness, and time show the subtle variations and traits present in the melanoma dataset. Understanding the underlying patterns and trends in the course of melanoma and the demography of patients depends heavily on this kind of analysis.

Part (vii) - Insights generated from the data analysis and provide recommendations on aspects that should be investigated in more detail

Discussion of Insights from the Data

1. **Relationships Among Variables**: - The results of the research indicated that there was a minor decrease in the time to event or censoring when thickness or age increased. Weak negative correlations were found between time and both thickness and age. These correlations, however, do not appear to be prominent, suggesting that other factors may be more important in influencing the time to events in cases of melanoma. Older patients have a marginally higher probability of having thicker melanomas, according to the weakly positive connection found between thickness and age. This discovery may have to do with the accumulation of solar exposure over an extended period of time or delayed detection in the elderly.

2. **Gender Differences**: - Gender variations, significant in time and thickness, underline the criticality of incorporating gender in melanoma studies. (Rothman et al., 2008), as gender appears to be a significant factor influencing disease characteristics (Siegel et al., 2020).

3. **Distribution Characteristics**: - QQ-plots demonstrated that time and thickness do not follow a normal distribution perfectly, especially in their higher values (Wilk & Gnanadesikan, 1968). This departure from normalcy may have an effect on certain statistical studies and interpretations, indicating the necessity for non-parametric or more reliable techniques (Box & Cox, 1964).

Recommendations for Further Investigation

1. **Explore Additional Variables and Factors**: - Examine additional variables and circumstances, such as lifestyle choices, genetic predispositions, or particular treatment modalities, that may affect time, thickness, or age. Gaining insight into these extra elements may offer a more complete picture of the variables affecting the course of melanoma.

2. **Detailed Analysis Focusing on Gender Differences**: - A more thorough investigation concentrating on these distinctions is advised in light of the noteworthy results regarding time and thickness between genders. Gaining insight into the fundamental causes of these gender differences may help develop more specialized treatment plans and focused preventative initiatives.

3. **Longitudinal Analysis and Data Over Time**: - If longitudinal data are available, analyzing data over time may provide important insights into how melanoma progresses and what factors are associated with it. This would facilitate comprehension of the disease's progression and how different treatments affect it.

4. **Non-Parametric Methods for Non-Normal Distributions**: - Consider using non-parametric statistical techniques for variables that do not have normal distributions (Spearman, 1904). This method could yield more accurate insights and be more appropriate for analyzing skewed data (Tukey, 1977).

5. **Clinical Collaboration and Relevance**: - Work together with clinical specialists to interpret the results in a context related to medicine (Clark et al., 1989). Clinical decision-making methods should take patient age and melanoma thickness into particular into account (Balch et al., 2009).

6. **Further Research on Subgroups and Specific Patterns**: - The patient population may contain subcategories, according on the dataset. Examining these subgroups may provide more precise trends, protective variables, or distinctive features of the development of melanoma (McGovern et al., 1973).

The purpose of this thorough review and the recommendations that follow is to improve our knowledge of the traits and course of melanoma, laying the groundwork for further study and practical applications. By focusing on these crucial areas, future studies will possibly find critical insights that lead to improved patient outcomes and more effective melanoma treatments.

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